

Web Exercise 1: Introduction of GitHub and Online Data Science Resources

DUE Date: Posted on CANVAS.

Grade: 10 points

You can complete this exercise using your own notebook computers or desktops.

Estimated time to complete: **THREE hours**

GitHub Introduction (70 mins)

GitHub is a very popular software programming collaboration platform for source/version control. Many successful teams use GitHub to develop, share, and exchange their source code and software packages. The platform is widely used by start-ups, open-source projects, and even companies like Microsoft and Google. GitHub is also a very important resource for learning Data Science.

1. Read TWO Wikipedia Articles

- a. What is “Git”? <https://en.wikipedia.org/wiki/Git>
- b. What is “GitHub”? <https://en.wikipedia.org/wiki/GitHub>

(10 mins to read these definitions)

2. Watch TWO YouTube Videos

- a. “How GitHub works” (4 minutes)
<https://www.youtube.com/watch?v=w3jLU7DT5E>
- b. “GitHub Tutorial for Beginners” (20 minutes)
<https://www.youtube.com/watch?v=0fKg7e37bQE>

You may not understand all the content in this video when you watch it for the first time. That’s fine. Please go ahead and do Step 3 after watching this video. You can re-watch this video again after you complete the actual tutorial.



3. Please go to this website and follow their instructions for the GitHub Tutorial (40 minutes): <http://product.hubspot.com/blog/git-and-GitHub-tutorial-for-beginners>

When you install “git” in your local machine, please use all the default settings or selections during the installation, EXCEPT FOR THE FOLLOWING NOTES:

Note #1: When you create a new repo on the GitHub website, please use the following as the name of your new repo:

GEOG594-yournickname

or

BDA594-yournickname

Note #2: DO NOT check the box for “initialize this repository with a README” in this exercise. After creating the new repository, copy the command lines from “push an existing repository from the command line” section. For example:

```
git remote add origin https://GitHub.com/mingtsou/GEOG594.git
git push -u origin master
```

Hint #1: Before you make the Git commit, you may need to set up your username and email according to your GitHub registration. Please follow the two commands below to set up the global configuration:

```
git config --global user.email "you@example.com"
git config --global user.name "Your Name"
```

Hint #2: You can create a new branch on a local repo by using this command:

```
git checkout -b ming-new-branch
```

Key concepts in these tutorials:

- What is a “repo”?
- What are the popular commands in terminal modes (cd, cp, mv, ls, rm)?
- What are the differences between “pull” and “push”? What will “clone” do?

Additional Resources to learn Git and GitHub:

- “An Introduction to Git and GitHub by Brian Yu” on YouTube (40 mins, very nice introduction of some concepts): https://www.youtube.com/watch?v=MJUJ4wbFm_A
- Several companies have released “cheat sheets” for common Git commands:
 - GitHub’s Cheat Sheet:
<https://education.github.com/git-cheat-sheet-education.pdf>
 - Atlassian’s Cheat Sheet:
https://www.atlassian.com/dam/jcr:e7e22f25-bba2-4ef1-a197-53f46b6df4a5/SWTM-2088_Atlassian-Git-Cheatsheet.pdf
 - GitLab’s Cheat Sheet:
<https://about.gitlab.com/images/press/git-cheat-sheet.pdf>
- This article that describes “GitHub flow.” It describes a simple workflow for using Git with a team: <https://guides.github.com/introduction/flow/>
- The blog post “A successful Git branching model” by Vincent Driessen. It describes a more detailed workflow for large projects with long release periods:
<https://nvie.com/posts/a-successful-git-branching-model/>
- There is detailed documentation for each individual git command here:
<https://git-scm.com/docs>

Markdown language (25 mins)

4. Now you need to learn how to create the README.md file in your new GitHub Repo (BDA594-yourname). You will also need to learn “Markdown” language to edit your README file.

In your newly created GitHub Repo, click the green “Add a README” button. Now you will be in Edit mode for your README file. Notice that the README file is written in Markdown language. Please read the introduction about Markdown in this link:

<https://guides.GitHub.com/features/mastering-markdown/>

After learning the Markdown style, please edit your README.md file to include the following content:

1. Your First name and Last name
2. The URL to the BDA/GEOG594 class (in CANVAS)
3. Your own definition of Big Data
4. (Optional) One photo. Make sure that you own the copyright of the photo.

Print out this README.md into a PDF file. Make sure you print from GitHub's regular display, not from inside the editor. This PDF will be one of your lab exercise submission items.

Create a GitHub Website (25 mins)

5. Every user can create a free website using GitHub. Please follow the instructions [HERE](https://guides.github.com/features/pages/): <https://guides.github.com/features/pages/>

Create a new repo using "yourname.github.io." By following the instructions, you will create your own website hosted by GitHub. Copy the link of your GitHub website for the lab report. (<https://##yourname###.github.io/>). Please introduce your selected data set and data science book on your GitHub Website.

Note: In the GitHub Setting, --> Page --> Under the Branch --> Select "master" (do not select None). --> Save.

Resources for Learning Data Science (30 mins)

6. Read the article "Top Data Scientists to Follow & Best Data Science Tutorials on GitHub" first: <https://www.analyticsvidhya.com/blog/2015/07/GitHub-special-data-scientists-to-follow-best-tutorials/>
7. Go to the awesome-datascience GitHub: <https://github.com/bulutyazilim/awesome-datascience>. From the "Datasets" list, select one data set that interests you. Introduce the data set on your website. Make sure you include its URL. Also describe some potential applications for the data set.
8. Got to the free-data-science-book GitHub: <https://github.com/chaconnewu/free-data-science-books>. Select one book from the list that interests you. Introduce the book on your website. Make sure you include its URL and its **author/institute** information. Also explain briefly why you are interested in this book.

Additional Resources for Git Code Sharing and Management

- GitLab: <https://about.gitlab.com> GitLab is a tool for building web-based Git repo systems from a centralized server (You can build your own "GitHub" site).
- Bitbucket: <https://bitbucket.org> Similar to GitHub. Bitbucket focuses on services to professional developers with private proprietary software code. Free private repos with a small team (up to 5 users).

After finishing these exercises, **please use your own words to answer the following questions (next page): (DO NOT COPY any web resources or Wikipedia texts. We will check your answers with CANVAS Turnitin tools to verify that your responses are uniquely yours.)** By submitting your answers (paper) to CANVAS, you agree: (1) that you are submitting your paper to be used and stored as part of CANVAS User Privacy Policy. (2) that your institution may use your paper in accordance with your institution's policies.

AI Policy: Students should not use generative AI applications (such as ChatGPT) for **writing paragraphs or sentences of these questions and discussions.** Any use of generative AI outside of instructor-approved guidelines constitutes misuse. Misuse of generative AI is a violation of the course policy on academic honesty and will be reported to the Center for Student Rights and Responsibilities. **Any AI-assisted content must be clearly cited or attributed to your work in accordance with academic integrity standards.**

LAB-1 Questions:

1. What are the key functions of GitHub? Who are the users? Please use your own words to describe them (200 words – 300 words).
2. What are the differences between “clone” and “pull request” in GitHub? (Use your own words).
3. Introduce ONE selected “Dataset” from the [awesome-datascience] GitHub with its URL and describe its potential applications and values.
4. Introduce ONE selected “free-data-science-book” with its URL and **author/institute** information. Explain briefly why you are interested in this book.
5. Print out the README.md (in a regular display mode inside GitHub) as a PDF. Add this to your submission as Attachment #1 (at the end of the submitted PDF file).
6. Print out your own GitHub Website (###.github.io/) as a Web Page **from a Web Browser directly (such as Chrome) into a PDF document** (not from the Github web platform). Add this to your submission document as Attachment #2 (at the end of your PDF file). (Please make sure to introduce your selected data set and data science book on your GitHub Website).

Please submit your LAB-1 Answers (in a MS Word or a PDF file format with additional attachment included) to the CANVAS Assignment BEFORE the DUE DATE/TIME.

NOTE: Please combine multiple PDF or word files **into a single PDF or a Word file**, then submit it to the Assignment Submission in the CANVAS. (Here is the free online link to merge PDF files: <https://www.adobe.com/acrobat/online/merge-pdf.html>).